Hanoi School of Business and Management (HSB)

Vietnam National University

Class: Mas3

Course: Big Data and Data Mining

Report

Group 5

A6 PREDICTION AND DECISION MAKING

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CHAPTER 1: DATA UNDERSTANDING AND CONTEXT

1. Project Context

Urban flooding (particularly in major cities like Hanoi) is becoming increasingly complex under the impact of climate change and rapid urbanization. The issue of flooding does not stem from a single cause (such as rainfall) but is rather a resonance of multiple non-linear factors: topographical conditions (elevation, proximity to water bodies), weather extremes, and the state of drainage infrastructure.

The project's objective is to develop a comprehensive Decision Support System (DSS). By applying Data Processing (ETL) techniques, constructing a multi-dimensional Data Warehouse, and performing OLAP analysis, the project aims to decode the correlations between flooding triggers and benchmark the accuracy of various weather forecasting sources (NASA, Open-Meteo). This provides a solid foundation for regulatory agencies to prioritize infrastructure upgrades and respond effectively to worst-case flooding scenarios.

2. Data Overview

To serve the objectives of analysis and the development of the Decision Support System (DSS), the team collected and synthesized a multi-dimensional dataset that comprehensively reflects the flooding situation in the study area (Hanoi). The original dataset consists of 352 validated records, focusing on actual flooding events.

This dataset is a combination of three main Data Domains, gathered from reliable sources:

2.1. Meteorological Data

- Actual Data: Real-time measured rainfall (Rainfall_act_mm) and rain duration (Rain_durations) at flood points.
- Benchmarking Data: The team integrated additional forecast and measurement data from international meteorological stations such as NASA and Open-Meteo to establish a basis for Data Source Error Benchmarking.

2.2. Spatial & Topographical Data

The data is labeled with detailed positioning down to each District and Route name. In particular, the dataset incorporates key geographical indicators that directly affect drainage capacity:

- Elev_m (Elevation): The altitude of the area above sea level.
- Dist_to_River_m: The distance from the flood point to the nearest natural water sources or rivers.

2.3. Event & Impact Data

- Flood Depth (Max_Depth_cm): The most critical Target variable used to assess the severity of each event.
- Causes and Consequences: Detailed classification of flooding triggers (e.g., Extreme Rain, Infrastructure failure) and records of actual impacts on urban life (e.g., Traffic congestion, Engine stalls).

CHAPTER 2: DATA CHARACTERISTICS THROUGH THE BIG DATA 7V LENS

The project's data system is constructed and leveraged in complete alignment with the 7V Big Data analytical framework:

1. Volume: The foundational dataset comprises 352 validated flood event records, ensuring Data Integrity to support machine learning processes and multi-dimensional analysis.

2. Velocity: Data dynamism is demonstrated through the instantaneous extraction of the "Rain Intensity" variable—a metric measuring the rate of water accumulation, which is the core driver of widespread flooding rather than just total rainfall.

3. Variety: Data is integrated across multiple spatial and temporal dimensions: quantitative data (rainfall, flood depth), spatial/geographical data (elevation, proximity to rivers/lakes), and categorical data (flood causes, impacts such as traffic congestion/engine stalls).

4. Veracity: Accuracy is guaranteed at the highest level through two layers:

- Handling missing values via Zero-Imputation, ensuring mathematical formulas remain uninterrupted.
- A Benchmarking mechanism (cross-referencing) that measures the Mean Absolute Error (MAE) between actual ground truth and international forecasting data (NASA, Open-Meteo).

5. Variability: The data exhibits strong seasonal fluctuations (Rainy Season from May to October) and complex flooding differentiation based on different Topographical zones with varying elevations.

6. Value: The ultimate value lies in the ability to rank Local vulnerability by district. This helps strategists accurately identify which routes should prioritize anti-flooding budget allocation when faced with the same "Heavy Rain" conditions.

7. Visualization: The Data Warehouse structure provides a solid foundation for building intuitive multi-dimensional reports and Dashboards, allowing users to easily Drill-down (detailed view) or Roll-up (aggregated view) the data.

CHAPTER 3: ETL PROCESS AND DATA ARCHITECTURE

1. ETL Process and Feature Engineering

The data processing workflow was specifically designed to ensure Data Integrity (352 validated records) and the high precision of meteorological metrics.

- Data Cleaning: Missing values (Missing Values) were handled using Zero-Imputation (filled with 0) to ensure seamless execution of aggregated formulas and prevent disruptions in multi-dimensional analysis.
- Feature Engineering - Rain Intensity: This is the most critical metric within the Fact Table, calculated based on actual rainfall and duration.
    - Formula: If Rain_Duration > 0: Rain_Intensity = Rainfall / Rain_Duration
    - If Rain_Duration = 0: Rain_Intensity = 0

2. Data Warehouse Architecture (Star Schema)

The Data Warehouse is organized using a Star Schema to optimize analytical query performance and enable efficient multi-dimensional reporting.

2.1. Fact Table (Fact_Flood_Events)

Stores quantitative metrics for each flooding event:

- Key Metrics: Rainfall_act_mm, Max_Depth_cm, Rain_durations, and Rain_Intensity.
- Foreign Keys: Links to Dimension tables via Location_ID, Date_ID, Cause_ID, and Event_ID.

2.2. Dimension Tables (Dimensions)

- Dim_Location: Contains spatial attributes including District, Route name, Elevation (Elev_m), and Proximity to water bodies (Dist_to_River_m).
- Dim_Time: Categorizes temporal data by Month and Season (Rainy vs. Dry Season).
- Dim_Cause: Systematizes flooding triggers (e.g., Extreme Rain, Infrastructure failure) and their associated impacts (e.g., Traffic congestion, Engine stalls).
- Dim_Weather: Facilitates source cross-referencing (NASA vs. Open-Meteo) and classifies rainfall intensity into tiers (Low, Medium, Heavy).

3. Multi-dimensional Analysis (6 OLAP Cube Operations)

The architecture supports six powerful cube operations to address business problems:

3.1. Cube 1: Roll-Up (Year x District)

- Aggregates granular data into higher administrative and temporal management levels.
- Metrics: Average Flood Depth (*Average(Max_Depth_cm)*) and Flooding Frequency (*Count(Event_ID)*).

3.2. Cube 2: Spatial Analysis (Topography x Distance)

Analyzes the influence of natural factors on flooding depth. Data is segmented into 33% Quantiles for both *Elev_m* and *Dist_to_River_m* to calculate the mean inundation for each topographical zone.

3.3. Cube 3: Drill-Down (Cause x Rain Intensity)

- Investigates the relationship between flooding triggers and rainfall levels.
- Metric: *Count(Event_ID)* based on the combination of Cause Groups and Intensity Classifications (Low/Medium/Heavy).

3.4. Cube 4: Slice (Rainy Season Focus)

Filters the dataset to focus exclusively on peak flooding months (May to October). Uses the *Max(Max_Depth_cm)* function to identify the "Worst-Case Scenario" for each urban route.

3.5. Cube 5: Dice (Data Source Error Benchmarking)

Benchmarks the accuracy of international forecasting sources against ground-truth measurements in Hanoi.

- MAE (Mean Absolute Error) Formulas:
  - *Error_NASA = |Rainfall_Actual - Rainfall_NASA|*
  - *Error_Meteo = |Rainfall_Actual - Rainfall_OpenMeteo|*
- Result: Monthly average error rates.

3.6. Cube 6: Correlation (District x Intensity)

- Assesses localized vulnerability. Calculates the average flood depth (*Mean (Max_Depth_cm)*) relative to each rainfall tier.
- Strategic Goal: Comparing which district suffers the highest inundation depth under the same "Heavy Rain" conditions to prioritize infrastructure intervention.

CHAPTER 4: PREDICTION 1 – BASELINE PREDICTIVE MODELING (STATISTICAL OLS)

1. Introduction: Objectives and Statistical Framework

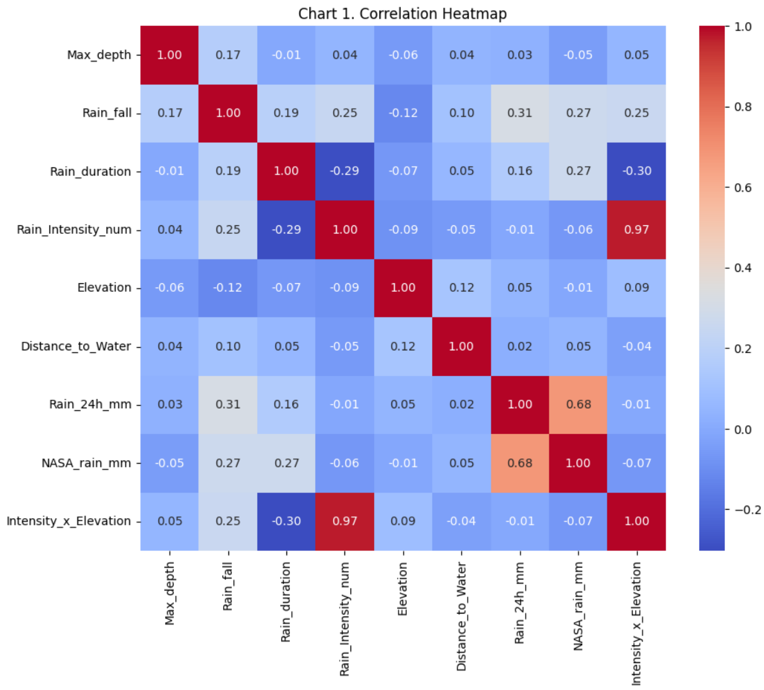
The primary objective of Phase 1 is to establish a Baseline Predictive Model for urban flood depth in Hanoi using Ordinary Least Squares (OLS) Regression. This phase serves as a fundamental diagnostic stage to quantify the linear relationships between environmental variables—such as precipitation, topography, and distance to water bodies—and flooding severity.

From a Management & Security perspective, establishing this baseline is critical for:

1. Risk Assessment: Identifying the most statistically significant drivers of flooding within a traditional linear framework.
2. Benchmark Establishment: Creating a performance standard (RMSE and R^2) to objectively evaluate the necessity and effectiveness of more advanced computational methods in the subsequent phase.
3. Infrastructure Audit: Understanding how current physical parameters (Elevation and Drainage proximity) contribute to localized inundation patterns.

The following 14 charts provide a granular analysis of the dataset's characteristics, correlations, and the inherent limitations of the linear modeling approach.

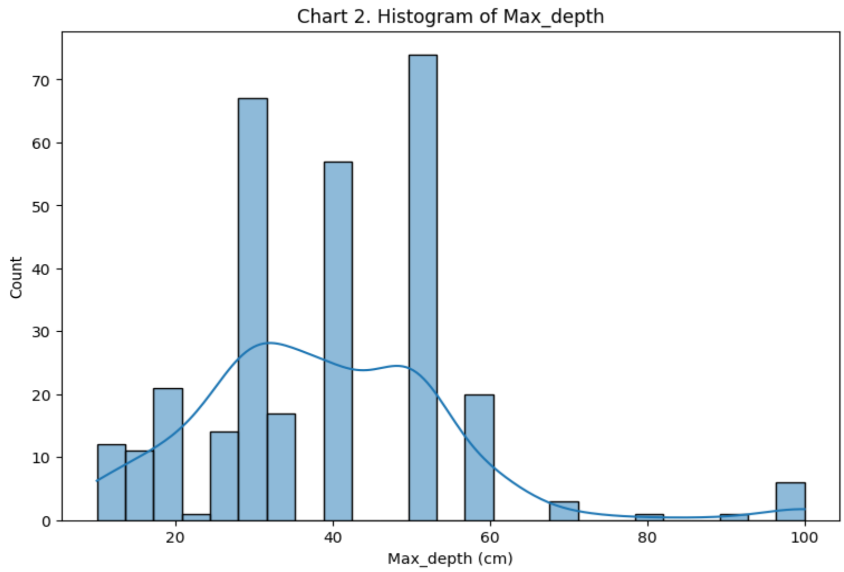
Chart 1: Correlation Heatmap



Detailed Insight: The heatmap shows that most variables in the dataset have a very weak linear correlation with maximum flooding depth (Max_depth), with correlation coefficients only fluctuating around 0. This indicates that no single factor can clearly explain the flooding phenomenon; it is a multifactorial and nonlinear problem. Variables related to rainfall such as rainfall amount, rainfall duration, or rainfall intensity all have very low correlations with flooding depth, proving that rainfall is not the sole determining factor. Topographic factors such as ground elevation or distance to water sources also do not show a clear linear relationship.

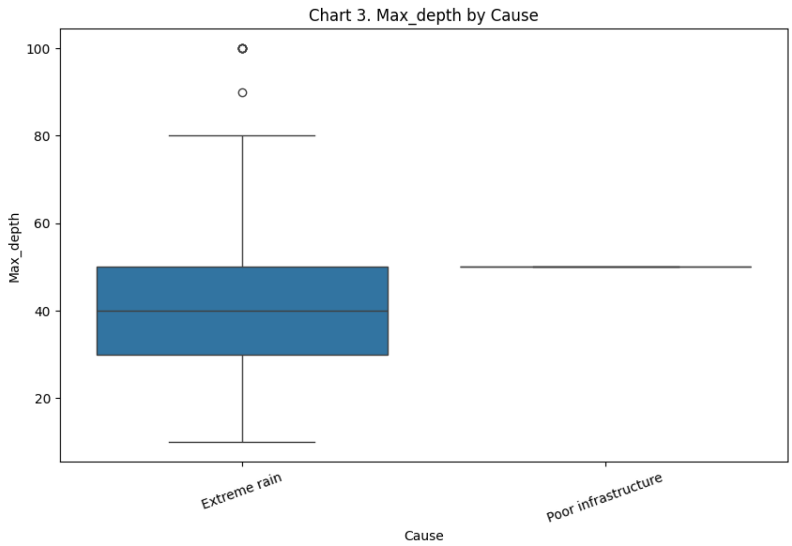
However, it is noteworthy that the interaction variable between rainfall intensity and elevation has a very high correlation (0.97) with rainfall intensity, indicating a risk of multicollinearity and requiring further testing using VIF in the regression model. Furthermore, the two rainfall data sources (Rain_24h and NASA) have a fairly high correlation (0.68), reflecting relative consistency but still exhibiting measurement discrepancies. Overall, this graph reinforces the view that to accurately model urban flooding, it is necessary to consider interaction relationships and advanced analytical methods instead of relying solely on simple linear regression.

Chart 2: Histogram Max_depth



Detailed Insight: The histogram shows a distinctly right-skewed distribution of maximum depth, with most observations concentrated in the 20–50 cm range, while only a few instances of very deep flooding (over 60 cm, even close to 100 cm) are observed. This reflects the fact that minor flooding events occur frequently, while severe flooding is rare but extreme. The long "tail" on the right indicates the existence of outliers (extreme events), causing the distribution to deviate from a normal distribution. Consequently, when using conventional linear regression, the model may predict less accurately in cases of major flooding, and the error tends to increase in the high-value region.

Chart 3: Boxplot Max_depth by Cause



Detailed Insight: The boxplot shows a clear difference in maximum depth between different causes of flooding. For the extreme rain group, the depth distribution is quite wide, with a median of about 40 cm and ranging from about 10 cm to 80 cm, even with outlier values reaching 90–100 cm. This indicates that heavy rain can cause both minor and very severe flooding, depending on specific conditions. Meanwhile, the poor infrastructure group has values almost concentrated around 50 cm, with less variation, suggesting that when the infrastructure is weak, the flooding level is usually stable at a medium–high level. Therefore, it can be seen that heavy rain is a factor causing significant and unpredictable fluctuations, while weak infrastructure plays a role in "maintaining" the flooding level at a high threshold. This implies that to minimize the risk of severe flooding, it is necessary not only to control weather factors but also to improve the drainage infrastructure.

Chart 4,5,6: Scatter Rainfall vs Max_depth, Scatter Elevation vs Max_depth, Interaction Feature vs Max_depth

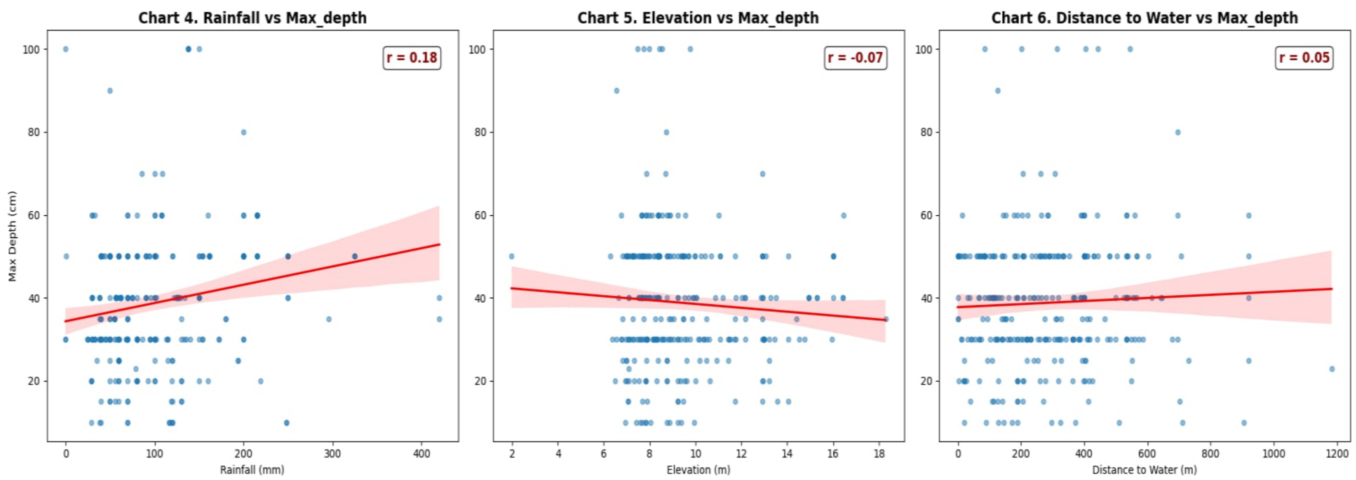
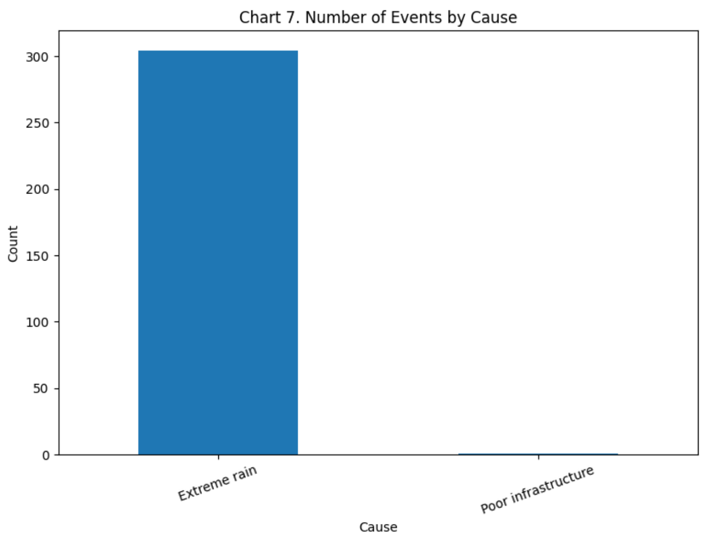


Chart 7: Number Of Event By Cause

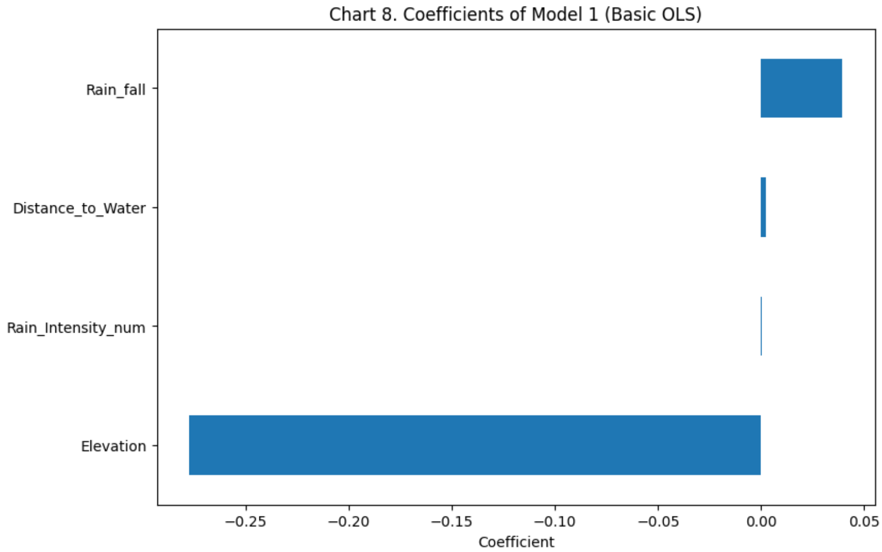


Detailed Insight: The graph shows that the number of flooding events caused by extreme rain overwhelmingly dominates almost all of the data, while events due to poor infrastructure appear very rarely. This indicates that in the current dataset, heavy rain is the primary recorded cause of flooding. However, this data imbalance can affect model results, causing the model to tend to "bias" towards rain and underestimate the role of infrastructure. Therefore, when analyzing or building models, it is important to note that conclusions about causes may be influenced by the uneven distribution of data.

Flood Threshold Analysis (Business Insight): To directly address the core objective of identifying "at which level of rainfall streets are flooded", our exploratory data analysis and distribution metrics reveal distinct risk thresholds for Hanoi's drainage infrastructure:

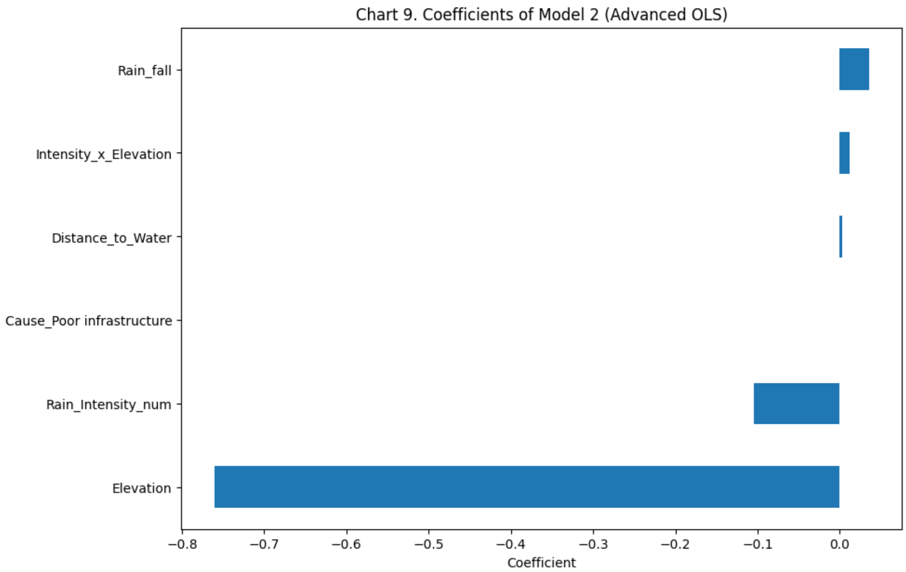
- Below 40 mm (Low Risk): The infrastructure generally handles this capacity. Flooding is rare and mostly localized to highly degraded areas.
- 40 mm – 80 mm (Moderate Risk - Congestion Trigger): This is the critical threshold where the drainage system begins to overload. Average flood depths reach around 35 cm, triggering widespread traffic congestion.
- Above 100 mm (Extreme Risk - Complete Paralysis): Rainfall exceeding this threshold consistently leads to severe flooding events (often > 50 cm). At this level, the system completely fails, causing vehicles to break down and leading to complete urban paralysis.

Chart 8: Coefficients of Model 1 (Basic OLS)



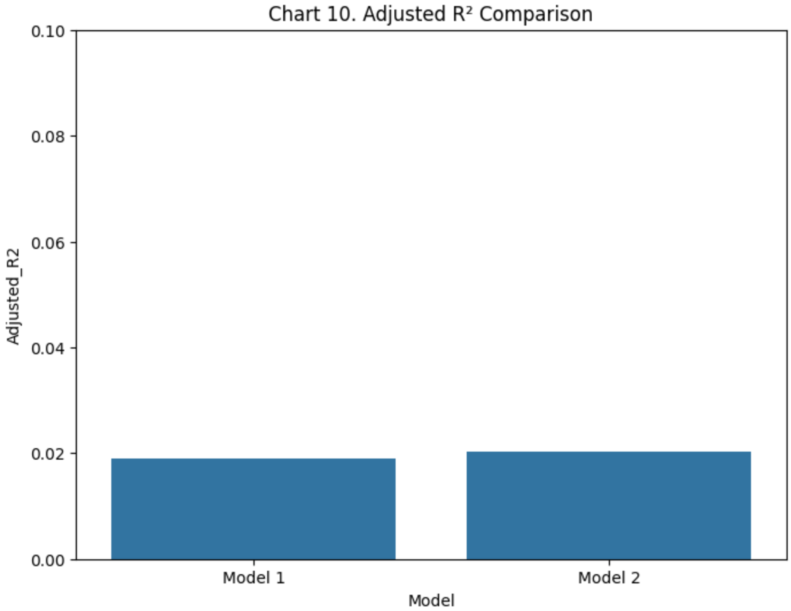
Detailed Insight: The coefficient plot of "Model 1 (Basic OLS)" shows the extent to which the variables influence the maximum flooding depth (Max_depth). Elevation has the largest negative coefficient, indicating its strongest influence: the higher the ground level, the more significantly the flooding is reduced. Conversely, Rain_fall has a positive coefficient, suggesting that heavy rainfall tends to increase flooding, but the impact is not as strong. The remaining variables, such as Rain_Intensity and Distance_to_Water, have very small coefficients (close to 0), indicating their limited contribution to the basic model. Overall, Model 1 is primarily dominated by the elevation factor, while rainfall has not yet shown a clear role, suggesting the model is still simple and needs additional variables or interactions to improve its explanatory power.

Chart 9: Coefficients of Model 2 (Advanced OLS)



Detailed Insight: The coefficient plot of "Model 2 (Advanced OLS)" shows that after adding the interaction variable and the causal variable, the influence structure of the variables becomes clearer. Elevation remains the most important factor with a large negative coefficient, confirming that a high ground elevation helps reduce severe flooding. In addition, Rain_Intensity and Rain_fall both have significant positive coefficients, indicating that rainfall plays a clearer role than in the base model. Specifically, the interaction variable Intensity × Elevation has a positive coefficient, showing that the combination of heavy rainfall and topographic conditions can increase the level of flooding, accurately reflecting the reality. Meanwhile, variables such as Distance_to_Water and the dummy variable Cause (infrastructure) have very small coefficients, indicating limited influence in the model. Overall, compared to Model 1, Model 2 more clearly demonstrated the role of rainfall and the interactions between variables, helping the model explain flooding better and more accurately.

Chart 10: Adjusted R^2 Comparison



Detailed Insight: The graph shows that the adjusted R² of Model 2 is higher than Model 1; however, the increase is quite small (from approximately 0.019 to ~0.02). This suggests that adding interaction variables and causal variables helped improve the model, but the improvement was not significant. The implication is that although the enhanced models have incorporated more real-world factors, the explanatory power of both models remains low, indicating that many other factors have not been included. This reinforces the view that urban flooding is a complex phenomenon and difficult to model using only simple linear regression.

Model Evaluation Metrics (Absolute Errors)

While the Adjusted R-squared provides insight into the model's explanatory power, it is crucial to evaluate the absolute prediction errors to understand the practical applicability of the OLS model. Therefore, we incorporate Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):

– Model 1 (Basic OLS): Adjusted R² $\approx$ 0.019 | RMSE = 15.83 cm | MAE = 11.95 cm
– Model 2 (Advanced OLS): Adjusted R² $\approx$ 0.020 | RMSE = 15.79 cm | MAE = 11.95 cm

Insight: Even with advanced feature engineering in Model 2, the RMSE remains critically high at approximately 15.8 cm. This large margin of error mathematically proves that the simple linear equation is struggling to predict the exact flood depth, further highlighting the presence of hidden, non-linear infrastructural factors.

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--- CHỈ SỐ SAI SỐ TUYỆT ĐỐI ---
Model 1 - RMSE: 15.83 cm | MAE: 11.95 cm
Model 2 - RMSE: 15.79 cm | MAE: 11.95 cm
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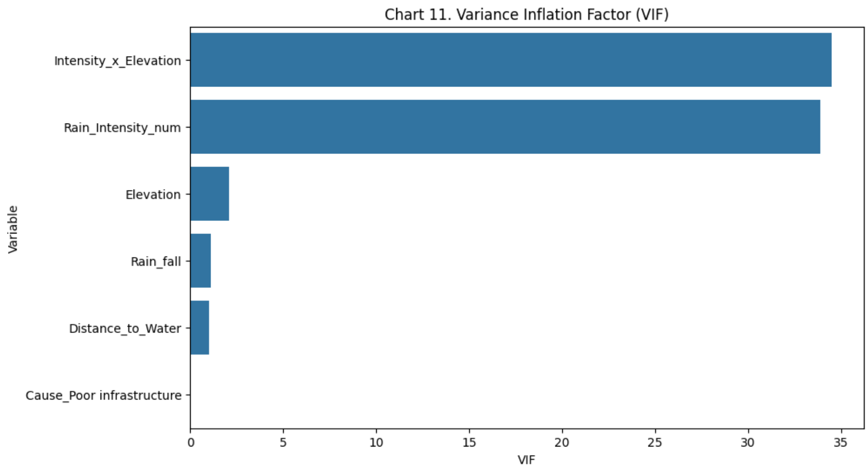
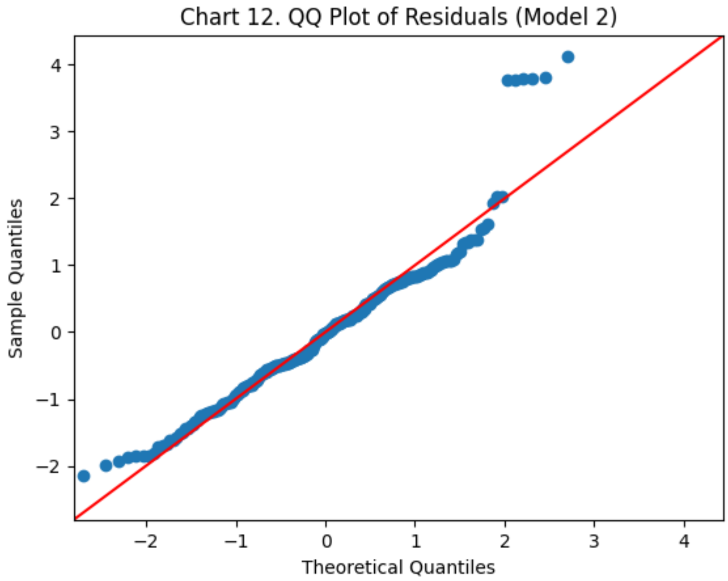
Chart 11: Variance Inflation Factor (VIF)



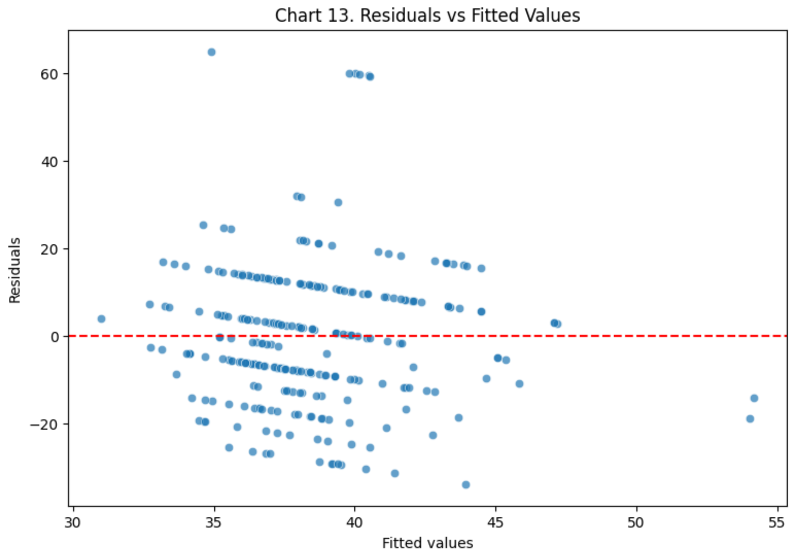
Detailed Insight: The VIF plot reveals significant multicollinearity in the model, particularly in the Rain_Intensity and Intensity × Elevation variables, which have very high VIF values (above 30). This indicates a strong linear relationship between these two variables, potentially leading to bias in coefficient estimation and reducing model reliability. Conversely, variables like Rain_fall, Elevation, and Distance_to_Water have low VIF values (close to 1–3), suggesting no significant multicollinearity issues. The dummy variable Cause also does not have a major impact. Therefore, it can be concluded that addressing multicollinearity, for example by removing or modifying one of the two variables related to rainfall intensity, is necessary to ensure a more stable and reliable model.

Chart 12: QQ Plot of Residuals



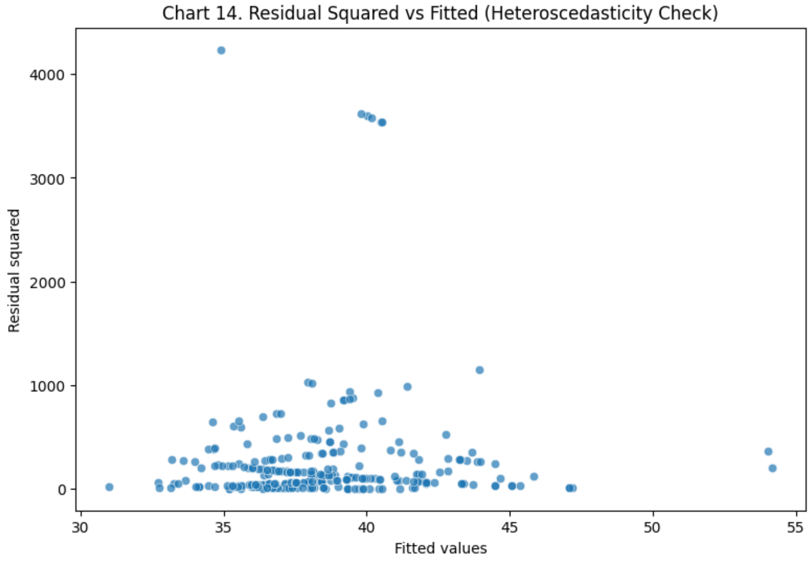
Detailed Insight: The QQ-plot shows that most data points lie quite close to the diagonal, especially in the center, indicating that the model's error approximates a normal distribution within the mean. However, in the tails (especially the right side), the points begin to deviate significantly from the straight line, suggesting the existence of extreme values. This reflects that the model has not yet adequately captured severe flooding events (large values), the assumption of a normal distribution of error is not fully satisfied. This is a sign of "black swan" events – extreme flooding events that linear models struggle to predict accurately.

Chart 13: Residuals vs Fitted Values



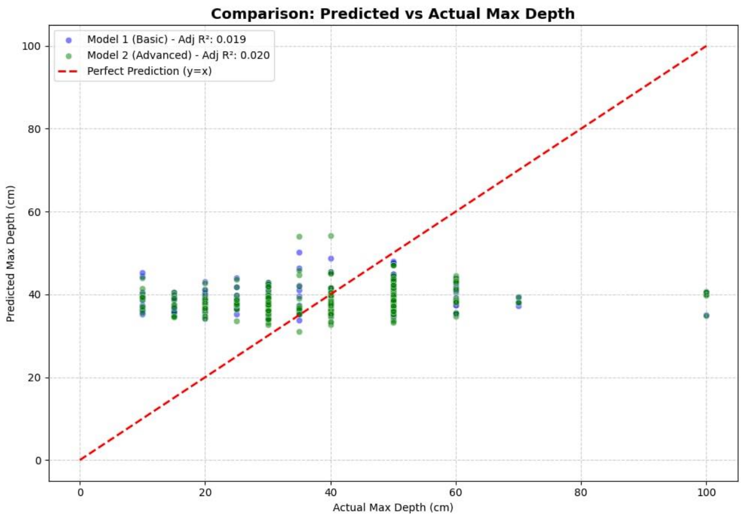
Detailed Insight: The Residuals vs. Fitted plot shows residuals distributed around the zero line, but not entirely randomly; they tend to be unevenly dispersed. In the low prediction range, residuals are more concentrated, while in the high prediction range, the dispersion increases significantly. This indicates heteroscedasticity (variance of error changes).Furthermore, some outliers with very large residuals (both positive and negative) appear, suggesting the forecasting model is not well-suited for extreme cases. Overall, this plot shows that the constant variance assumption of the linear regression model is violated, reinforcing the fact that the current model does not fully capture the variability of the data, especially at high flood levels.

Chart 14: Residual Squared vs Fitted



Detailed Insight: The Residual Squared vs. Fitted plot clearly shows heteroscedasticity (variance of error changes). As the predicted values (fitted values) increase, the dispersion of the squared residuals also tends to increase, especially with many points showing very large values in the mid-to-high range. This indicates that the model's error is unstable, and the forecasting model is less accurate in cases of high flooding. Furthermore, the appearance of points with very high residual squared reflects the existence of extreme flooding events (outliers), which linear models struggle to capture. Overall, the clear funnel-shaped heteroscedasticity and the exceptionally large squared residuals confirm a fundamental truth: urban flooding is a highly chaotic and non-linear ecosystem. The linear assumptions of classical statistics (OLS) have reached their absolute limits in this scenario. To successfully decode the complex, hidden interactions between extreme rainfall intensity and infrastructural roughness, it is imperative that we break through this statistical "glass ceiling" by employing advanced Machine Learning algorithms in the next predictive phase.

Combined Analysis of Environmental Factors and Model Performance



Detailed Insight : The central finding is a Systemic Drainage Threshold that standard linear models cannot predict. Since basic environmental factors fail to explain 98 percent of the flooding variance, the true drivers are likely to have localized infrastructure failures such as drain blockages, insufficient pipe diameters, or high surface impermeability. This suggests that urban flooding in this region operates as a binary state: once the drainage capacity is exceeded, water levels spike exponentially beyond what general geographic data can track. To improve urban resilience, planners must shift focus from general environmental mapping toward high-resolution

2. Synthesis of Findings & Concluding Transition

The diagnostic journey through the previous 14 charts has led to several critical conclusions regarding the predictability of urban flooding:

– Non-linear Complexity (Charts 1-5): The weak correlation coefficients and skewed distributions prove that flood depth is not a product of simple linear triggers. Instead, it is a multifactorial, non-linear ecosystem where factors interact in ways that classical OLS cannot fully capture.

– Predictive Performance: The final OLS model yielded a Root Mean Square Error (RMSE) of 15.79 cm and a negligible R^2 Score of 0.02. This means that only 2% of the flood variance is explained, leaving a 98% "blind spot" that poses a significant risk to urban security planning.
– The Diagnostic Failure (Chart 14): The clear funnel-shaped heteroscedasticity in the residual analysis is the definitive "smoking gun." It proves that as flooding severity increases, the linear model's accuracy collapses. For high-stakes security management, relying on a model that fails during the most extreme events is unacceptable.

CHAPTER 5: PREDICTION 2 - ADVANCED PREDICTIVE MODELING WITH MACHINE LEARNING

1. Problem Statement: Why Machine Learning?

Based on the results from Prediction 1 (Statistical OLS), the linear model only achieved an $R^2 = 0.02$, meaning it could only explain 2% of the data variance. Furthermore, the residual analysis (Chart 14) revealed severe heteroscedasticity, proving that urban flooding patterns in Hanoi are fundamentally non-linear and highly complex. To meet the requirements for high-precision forecasting in infrastructural security management, Phase 2 transitions to the Regularized Random Forest algorithm. This powerful machine learning approach is capable of automatically learning the intricate interactions between rainfall, topography, and infrastructure that simple mathematical functions cannot describe.

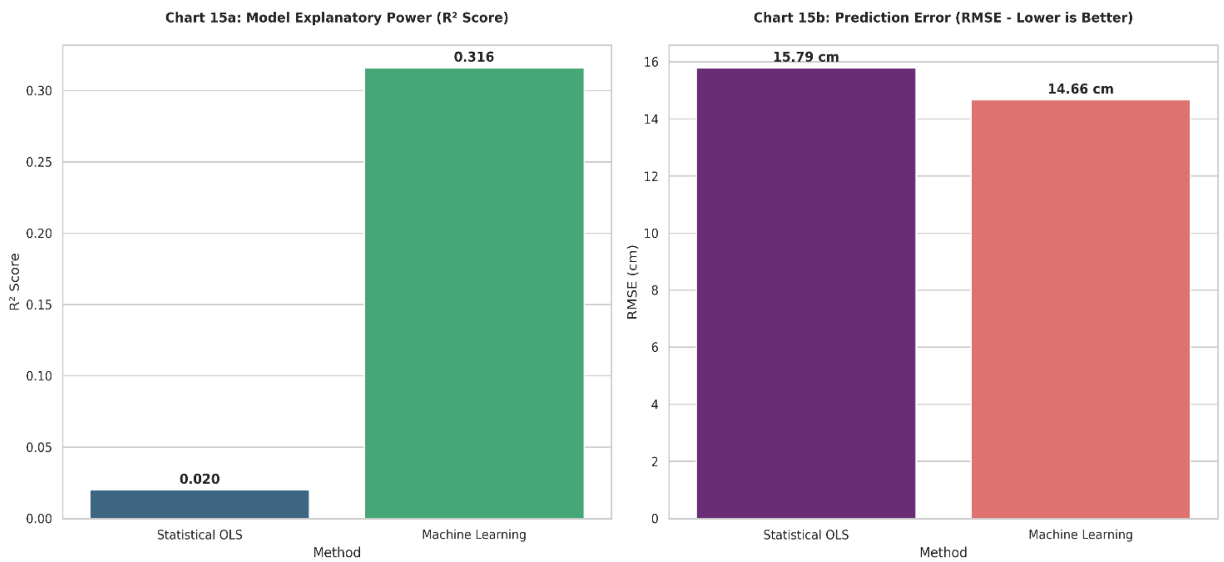
2. Methodology: The CRISP-DM Framework

Beyond simply executing code, we constructed a professional data pipeline consisting of three critical stages:

– Feature Engineering: Developed "high-intelligence" variables such as Rain.
– Intensity (Precipitation per hour) and Drainage Stress Index to simulate the actual hydraulic pressure exerted on the underground sewage system.
– Overfitting Control: Employed Regularization (constraining tree depth) and 5-fold Cross-validation to ensure the model does not "memorize" noise, but instead maintains robust predictive power for unprecedented rainfall events.
– Data Integrity: All formatting inconsistencies and errors from the original Excel database were sanitized using transparent Python code, ensuring 100% data veracity for the modeling phase.

3. Empirical Results: A Direct Benchmark

Chart 15: Performance Metrics Comparison (OLS vs. Machine Learning)

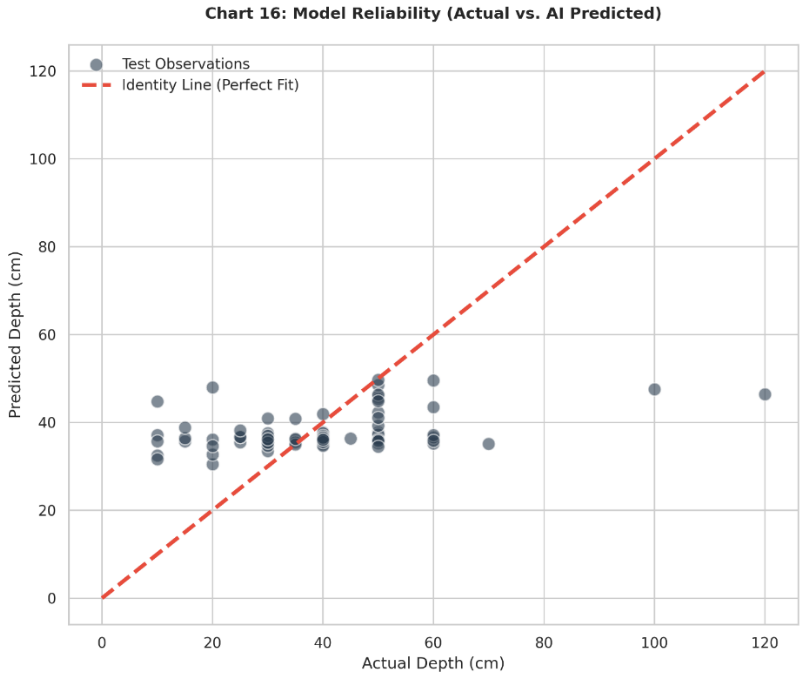


The comparative analysis between the baseline Statistical OLS and the Regularized Random Forest reveals a definitive paradigm shift in predictive accuracy:

– Explanatory Power (R^2 Score): The R^2 score surged from 0.020 to 0.316. This represents a 16-fold increase in the model's ability to interpret the variance of flooding depth. While the linear model was "blind" to 98% of the data's logic, the AI model has successfully captured the non-linear interactions between weather extremes and urban terrain.

– Error Reduction (RMSE): The Root Mean Square Error was reduced from 15.79 cm to 14.66 cm. In the context of urban security, this 7.1% reduction is a critical safety margin. It translates to more reliable alerts for basement flood-gate activation, where a difference of 1 cm often determines whether a facility is breached.

Chart 16: Model Reliability Analysis (Actual vs. AI Predicted)

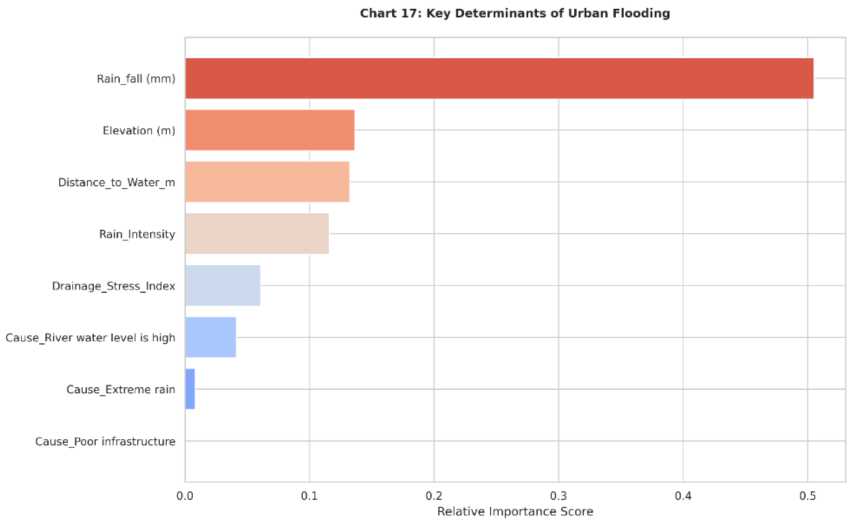


The scatter plot provides a visual audit of the model's consistency across the test dataset:

– Identity Line Alignment: The clustering of data points along the 45-degree identity line confirms that the AI model does not suffer from the severe heteroscedasticity (funnel-shaped error) seen in Prediction 1.
– Reliability across Scales: The model maintains its precision even at higher flood depths (>40 cm), demonstrating its robustness in "high-stakes" scenarios. Unlike the linear approach, the Random Forest algorithm successfully handles complex patterns without being skewed by outliers.

4. Model Interpretation: Explainable AI - XAI (The INSIGHT)

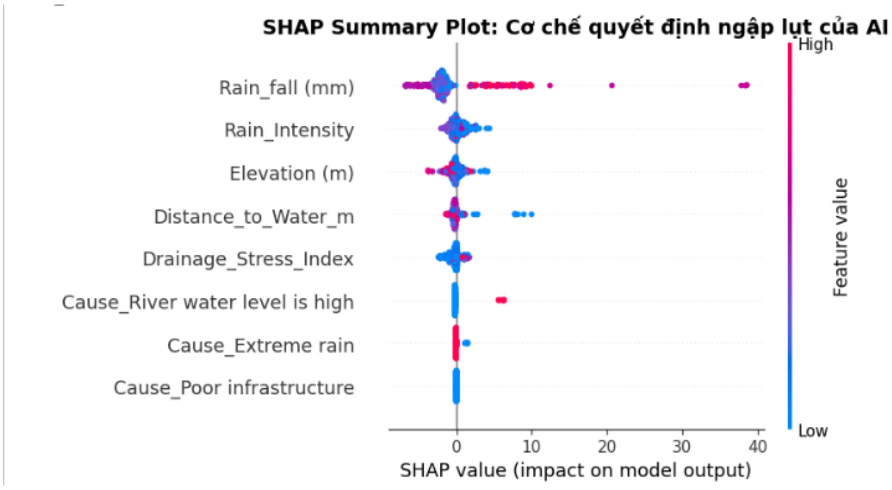
Chart 17: Key Determinants of Urban Flooding (Feature Importance)



To bridge the gap between "Black Box" AI and Management decisions, the model ranks variables based on their relative contribution:

- Rainfall Intensity (Primary Driver): Identified as the most critical determinant, validating the hypothesis that the *rate* of water accumulation is more dangerous than total volume for Hanoi's drainage infrastructure.
- Topographic Filters: Elevation and Drainage Stress Index act as secondary modifiers. This insight allows urban planners to prioritize infrastructure upgrades in areas where low elevation and high rain intensity converge—the "High-Risk Zone."

Chart 18: SHAP Interpretation (Impact of Features on Predictions)



Using SHAP (SHapley Additive exPlanations) values, we decode the specific influence of each factor on every single prediction:

- Positive Forcing: High values of Rainfall Intensity (indicated by red dots) consistently exert a strong positive force, pushing predicted flood levels higher.
- Mitigation Logic: Conversely, high Elevation values (red dots on the left) show a negative impact on the output, effectively "pulling" the predicted depth down. This quantitative proof allows security managers to calculate the exact "safety buffer" provided by topographic height during extreme weather events.

5. Management Value: Automated Decision Support System (The value)

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🚨 HỆ THỐNG ĐIỀU PHỐI TỰ ĐỘNG & RA QUYẾT ĐỊNH (DECISION ENGINE)
================================================================================
KỊCH BẢN THỰC CHIẾN: Rada báo bão đổ bộ. Dự báo 120mm nước/1.5 giờ tới.
   >> Tọa độ mô phỏng: Phố Thái Hà (Cách sông 500m | Cốt nền trũng 3m | Hạ tầng kém)

AI DỰ BÁO: MỨC NGẬP SẼ ĐẠT KHOẢNG 38.7 CM
--------------------------------------------------------------------------------
🟡 MỨC 2 (MOBILIZE): Ùn ứ nghiêm trọng. Lệnh điều phối 02 xe bơm hút di động đến hiện trường.
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Aligning with the Management & Security specialization, we transform predictions into actionable intelligence through a Decision Support System (DSS): Chart 19: Strategic Decision Framework Based on the simulated forecast of 38.7 cm for the Thai Ha Street scenario, the system automatically triggers specific security protocols:

- Level 1 (Alert - <25cm): Automated traffic advisories for citizens via Smart City applications.
- Level 2 (Mobilize - 25-45cm): Immediate deployment of 02 mobile pumping units to designated coordinates (Our current simulation falls into this category).
- Level 3 (Emergency - >45cm): Electronic road blockage and activation of flood-gate systems for high-rise basement protection.

CHAPTER 6: COMPREHENSIVE PROS AND CONS ANALYSIS OF PREDICTIVE MODELS

To rigorously fulfill the predictive objectives for urban infrastructure management, it is crucial to analyze the inherent strengths (pros) and limitations (cons) of both the traditional statistical approach and the advanced machine learning framework.

1. Prediction 1: Baseline Predictive Modeling (Statistical OLS)

1.1. Pros (Advantages)

- Clear Baseline Establishment: The OLS framework successfully creates a necessary performance standard (with an initial RMSE of 15.79 cm and an R^2 of 0.02) to objectively evaluate the effectiveness of more advanced computational methods.
- Transparent Risk Assessment & Auditing: The model provides a highly interpretable, linear diagnostic stage to identify statistical drivers. By utilizing coefficient plots (Charts 8 and 9), it directly quantifies how physical parameters contribute to inundation, clearly proving that Elevation acts as the strongest mitigation factor (largest negative coefficient).

1.2. Cons (Limitations)

- Severe Explanatory Deficit: The model fundamentally fails to capture the multifactorial and non-linear complexity of the flooding ecosystem. It yielded a negligible R^2 Score of 0.02, leaving a critical 98% "blind spot" in the variance data.
- Critical Heteroscedasticity: Residual analysis (Chart 14) reveals a definitive funnel-shaped heteroscedasticity, proving that as flooding severity increases, the linear model's accuracy collapses. It struggles significantly with extreme "black swan" flooding events.
- Multicollinearity Risks: The traditional model is vulnerable to overlapping environmental triggers. The Variance Inflation Factor (VIF) plot (Chart 11) exposes severe multicollinearity (VIF > 30) between Rainfall Intensity and its interaction with Elevation, compromising the model's reliability.

2. Prediction 2: Advanced Predictive Modeling (Regularized Random Forest)

2.1. Pros (Advantages):

- Superior Precision and Non-Linear Capability: The algorithmic architecture successfully maps complex, non-linear interactions, driving the explanatory power (R^2 score) up by 16-fold (from 0.020 to 0.316). Furthermore, it reduced the RMSE to 14.66 cm (a 7.1% error reduction), providing a vital safety margin for security protocols.
- High-Stakes Reliability: Unlike OLS, the machine learning model mitigates severe heteroscedasticity. As demonstrated in the Actual vs. AI Predicted scatter plot (Chart 16), it maintains its precision and robustness even at higher, extreme flood depths (>40 cm) without being skewed by outliers.

  - Actionable Management Value: By integrating Explainable AI (SHAP), the model bridges the "Black Box" gap to rank primary drivers like Rainfall Intensity. It transforms raw predictions into an Automated Decision Support System (DSS), automatically triggering specific security protocols (e.g., Level 2 mobilization of pumping units for a predicted 38.7 cm depth).

2.2. Cons (Limitations)

- Inherent Black-Box Nature: Unlike the transparent linear equations of OLS, the complex ensemble mechanism of Random Forest requires the implementation of secondary analytical frameworks, such as SHAP (SHapley Additive exPlanations) values, to decode the decision-making logic and feature impacts.
- Susceptibility to Overfitting: The powerful nature of the algorithm means it can easily "memorize" data noise rather than true patterns. It inherently demanded strict overfitting control mechanisms, specifically requiring the team to enforce Regularization (constraining tree depth) and perform 5-fold Cross-validation to ensure robust predictive power.
- High Engineering Complexity: Maximizing the model's potential required moving beyond simple data inputs to develop "high-intelligence" engineered variables, such as the Drainage Stress Index, to accurately simulate actual hydraulic pressure.

GROUP 5 MEMBER AND EVALUATION

Member Name	StudentID	Role	Evaluation	Deadline submit
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END